

Integration: PREP

Purpose

PREP (Price Repository) is an Oracle database hosted on Ørsted on-premises servers. PREP contains prices from third party data providers (published prices on commodity exchanges etc.) and internally constructed prices added by the Trading Team each day. It also contain prices for various "Risk" factors and currency exchange rates. The Pricing Tool needs all relevant prices from PREP to use as part of the pricing model calculations.

Method

Currently the required prices are pulled directly from PREP via a database link but with the new Pricing Tool this will change

There should be a manual upload trigger activated by a business user to fetch the data from PREP into the Pricing Tool.

Gas, Power, Exchange Rates and Risk add-on prices should also be able to be uploaded separately in case they change during the day and therefore the business wish to refresh those prices from PREP

Data source is Orsted SQL table dma_prep.DMA_PRICE_ALL_WV

Data structure:

product_bid	Identification of price curve
value_type_shortcode	
price_source_shortcode	Different for Prices, Price elements and Exchange rates
delivery_period_type_shortcode	Different for Prices, Price elements and Exchange rates
price_provider_shortcode	Different for Prices, Price elements and Exchange rates
notationtime	Which day are the prices for – dd-mm-yyyy
delivery_start	Validity from
delivery_end	Validity to
price	Value to be used
approved_time	Load date and time
Version	Load version – the view should only contains the last version per price curve, so taking max version is no longer needed
price_unit_shortcode	unit of measure

Pricing Tool configuration should allow for the definition of the different prices so that the parameters required to fetch the data from PREP are configured and can be changed by Orsted if needed

These parameters are ProductBid, PriceProviderShortName, PriceSourceShortName, ValueTypeName, DeliveryPeriodTypeShortName)

Prices are monthly granulated curves and published in EUR/MWh.

The curves must at any given time have the exact length of current year + five years starting with front month (i.e. between 60 and 71 months dependent on the time of year).

If the Pricing Tool requires a price for the current month then these should be created by the Pricing Tool by copying the value from the first future month.

Prices are found using the following logic:

```
SELECT
  a.product_bid,
  a.value_type_shortcode,
  notation_time,
  a.delivery_start,
  a.delivery_end,
  a.price,
  approved_time,
  a.version,
  a.price_unit_shortcode
FROM dma_prep.DMA_PRICE_ALL_WV a
WHERE
  a.PRODUCT_BID = <A>
  and a.VALUE_TYPE_SHORTNAME = <B>
  and a.DELIVERY_PERIOD_TYPE_SHORTNAME = <C>
  and a.PRICE_PROVIDER_SHORTNAME = <D>
  and a.PRICE_SOURCE_SHORTNAME = <E>
  and a.notationtime = <F>
```

The current configurations are below but this is not an exhaustive list

	A	B	C	D	E	F
Power Commodity	1577871	Bid	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1577871	Ask	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1577861	Bid	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1577861	Ask	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1571161	Bid	MO	DPR	SALES_TRADING	TODAY

Power Commodity	1571161	Ask	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1571171	Bid	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1571171	Ask	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1577831	Bid	MO	DPR	SALES_TRADING	TODAY
Power Commodity	1577831	Ask	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571601	Bid	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571601	Ask	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571481	Bid	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571481	Ask	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571491	Bid	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1571491	Ask	MO	DPR	SALES_TRADING	TODAY
Gas Commodity	1570591	Mid	MO	DPR	SALES_TRADING	TODAY
Power Exchange	1567553	Cls	AP	PRI	GTRM	TODAY-1
Gas Exchange	1573061	Is	MO	DPR	GTRM	TODAY-1
Price Element Power	1573071	TRAD1FEE	MO	DPR	SALES_PRICING	TODAY
Price Element Gas	1575451	ProfileFee	MO	DPR	SALES_PRICING	TODAY
Gas Monthly Product Example	1569961	CMDTYPrice	MO	DPR	SALES_PRICING	THIS month or most recent
Gas Monthly Product Example	1570431	Set	MO	DPR	SALES_PRICING	THIS month or most recent
Gas Monthly Product Example	1570001	CMDTYPrice	MO	DPR	SALES_PRICING	THIS month or most recent
Gas Monthly Product Example	1570421	Set	MO	DPR	SALES_PRICING	THIS month or most recent
Gas Monthly Product Example	1570031	TotalPrice	MO	DPR	SALES_PRICING	THIS month or most recent

Gas Monthly Product Example	1570031	Set	MO	DPR	SALES_PRICING	THIS month or most recent
Gas Monthly Product Example	1570011	CMDTYPrice	MO	DPR	SALES_PRICING	THIS month or most recent
Daily Product Example	1570021	TotalPrice	DY	DPR	SALES_PRICING	THIS month or most recent
Daily Product Example	1570021	Set	DY	DPR	SALES_PRICING	THIS month or most recent
Daily Product Example	1570441	Set	DY	DPR	SALES_PRICING	THIS month or most recent
Daily Product Example	1570461	Set	DY	DPR	SALES_PRICING	THIS month or most recent
Daily Product Example	1570171	Set	DY	DPR	SALES_PRICING	THIS month or most recent

Gas & Power Commodity Prices

If there are commodity prices in the database for PRICE_SOURCE_SHORTNAME = 'SALES_TRADING' use them.

If they are missing, try to pick up for the same parameters but with PRICE_SOURCE_SHORTNAME = 'SALES_PRICING'.

If those are also missing, raise an error.

The green certificate curves must always be the entire current year i.e if the contract starts in June 2024 then the prices are for the entire of 2024 and then forwards to the end of the contract

The price is monthly but will be the same price for the calendar year so for example if a contract is from June 2024 to December 2026 then each month in 2025 will have the same price and then each month in 2025 will have the same price etc

Exchange Rates

Published as EUR/DKK conversion.

Exchange Rate Currency curve is published on the last working day before today so TODAY-1 refers to 1 working day before TODAY i.e. On Wednesday it is from Tuesday and on Monday it is from Friday

If not found, find from the previous day i.e. TODAY-2 working days

If not then look for SALES_PRICING – all parameters are different in this case

If still not found raise an error

Currency curve contains data for today and the rest of this current month going forward

e.g if data taken on 23rd then data from 22nd to 31st and for months to 5 years and rest of curve year, one row per month

Take data from 1st of the following month to today – up to 5 years after this year

If 23/05/2024 it would from June to December 2024 and then up to Dec 2029

Price Elements

Price elements for Power are monthly granulated curves and are published in PREP in EUR/MWh.

Price elements for GAS are monthly granulated curves and are published in PREP in øre/m3.

There are many more price elements than those listed in the grid above

Price Examples

Price examples need to be imported to be included in the document templates.

Unlike prices and price elements, they do not come as curves but these are single month (for monthly products) and day (for daily products) values.

Monthly products: This should be the price to this current month or the most recent month

Daily products: This should be the price to this current month or the most recent month

Danish example prices are in øre/m3